23/10/2023

Wave Optics

1) destructive interference (dark)

path $= (2n+1)\frac{\lambda}{2}$ $\quad n = 0, 1, 2, 3 \ldots$

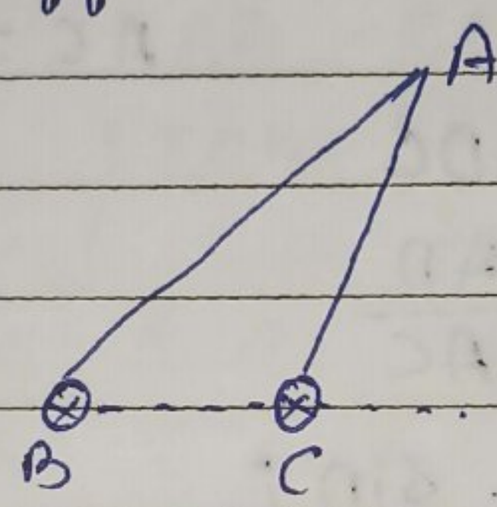
(odd $\times \lambda/2$)

2) constructive interference (bright)

path $= n\lambda = 2n \cdot \frac{\lambda}{2}$ $\quad n = 1, 2, 3 \ldots$

(even $\times \lambda/2$)

- Path difference



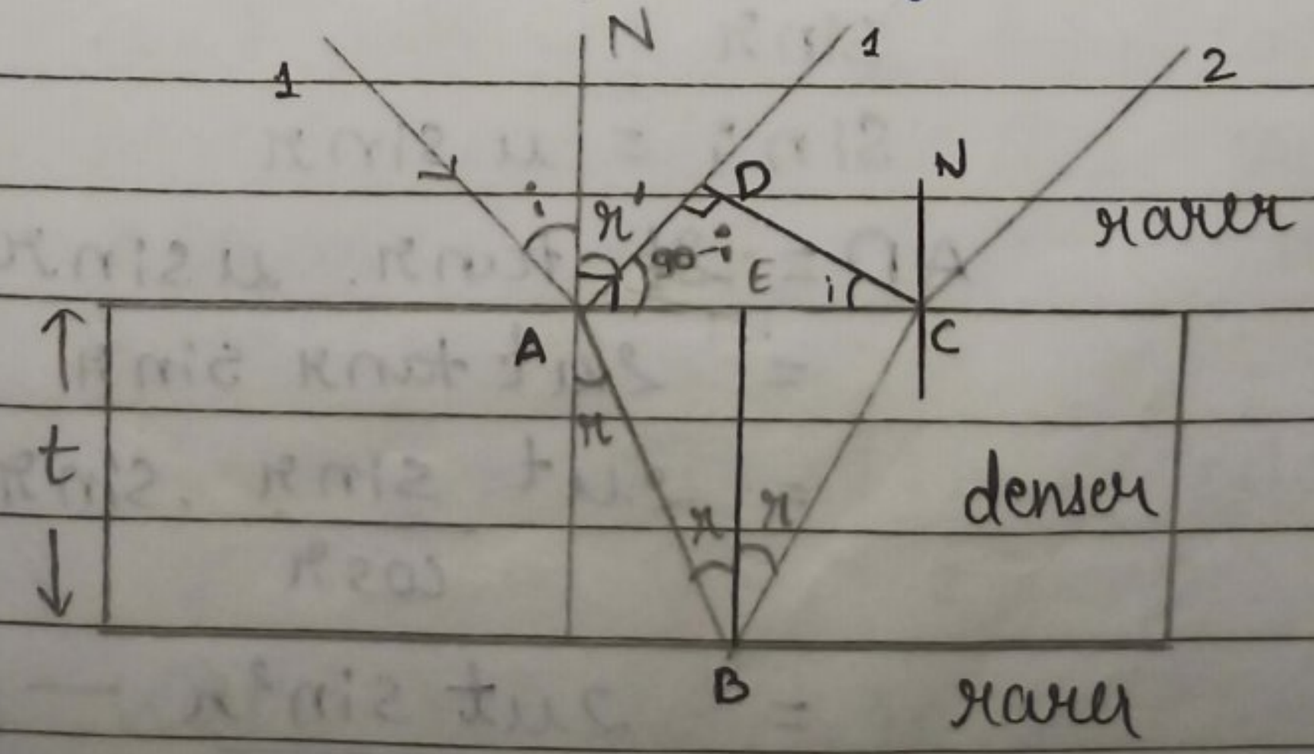
Geometrical path diff $= AB - AC$

optical path diff $= \mu(AB - AC)$

when lights reflects from denser surface additional path diff. of π gets added

* Interference of Light by thin film

(Reflected Ray) $\quad t = 0.5 - 10\,\mu m$



optical path diff $= \mu(AB + BC) - AD$ — (1)

In ΔADC,

$\angle A = 90 - i$ $\quad$ $EB = t$

$\angle D = 90^\circ$ $\quad$ t = thickness

$\angle C = i$

E = centre

$\therefore AE = EC$ $\quad$ $AC = AE + EC$

now in ΔADC

$$\sin i = \frac{AD}{AC}$$

$$AD = AC \cdot \sin i$$
$$= (AE + EC) \cdot \sin i$$
$$= 2AE \sin i \quad \text{— (2)}$$

now, ΔABE,

$$\tan r = \frac{AE}{EB} = \frac{AE}{t}$$

$$AE = t \cdot \tan r \quad \text{— (3)}$$

$$AD = 2t \cdot \tan r \cdot \sin i \quad \text{— (4)}$$

By snell's law

$$\frac{\sin i}{\sin r} = \mu$$

$$\sin i = \mu \cdot \sin r$$

$$AD = 2t \cdot \tan r \cdot \mu \sin r$$
$$= 2\mu t \tan r \sin r$$
$$= 2\mu t \frac{\sin r}{\cos r} \cdot \sin r$$
$$= \frac{2\mu t \sin^2 r}{\cos r} \quad \text{— (5)}$$

now ΔABC

$AB = BC$

$AB + BC = 2AB$ — (6)

now ΔABE,

$$\cos r = \frac{EB}{AB}$$

$$AB = EB/\cos r = t/\cos r$$

$\because AB = BC = \delta$

$\therefore$ from (1)

optical path difference $= \mu 2AB - AD$

$$2\mu (EB \cdot \cos r) - \frac{2\mu t \sin^2 r}{\cos r} = \frac{2\mu t}{\cos r} - \frac{2\mu t \sin^2 r}{\cos r}$$
$$= \frac{2\mu t}{\cos r}(1 - \sin^2 r)$$
$$= \frac{2\mu t}{\cos r}(\cos^2 r)$$
$$= 2\mu t \cos r$$

* For constructive interference (in case of their film)

Path diff $= n\lambda$

$$2\mu t \cos r - \frac{\lambda}{2} = n\lambda \quad [2\mu t \cos r - \lambda/2]$$

Actual path difference

$$2\mu t \cos r = nd + \frac{\lambda}{2}$$
$$= (2n+1)\frac{\lambda}{2} = (\text{odd})\frac{\lambda}{2}$$

where $n = 0, 1, 2, 3 \ldots$

$$\boxed{t = \frac{(2n+1)\lambda}{4\mu\cos r}}$$

t = thickness

$$\boxed{\therefore\ t = \frac{(2n+1)\lambda}{4\mu}}$$

$(r = 0 \quad \therefore \cos 0 = 1)$

* For destructive interference :-

$$\text{Path diff} = (2n+1)\frac{\lambda}{2}$$

$$2\mu t\cos r - \frac{\lambda}{2} = (2n+1)\frac{\lambda}{2}$$

$$2\mu t\cos r = 2n\frac{\lambda}{2} + \frac{\lambda}{2} + \frac{\lambda}{2}$$

$$2\mu t\cos r = n\lambda + \lambda = n\lambda$$

$[n = 1, 2, 3 \ldots]$

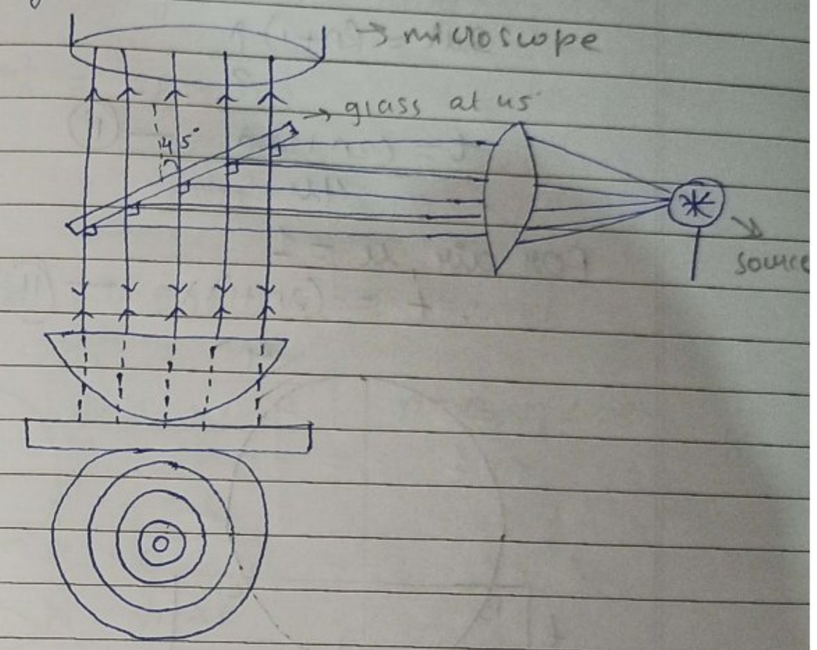
$$\boxed{t = \frac{n\lambda}{2\mu\cos r}}$$

$$t = \frac{n\lambda}{2\mu} \quad [\because r = 0]$$

* Newton's Ring

* Newton's Ring

- Experimental arrangement of Newton's rings.



* Expression for radius/diameter of Newton's Ring

→ Condition for destructive interference

$$2\mu t\cos r = n\lambda$$

for normal incident of light

$r = 0 \quad \therefore \cos r = 1$

$$\therefore 2\mu t = \cancel{\cos} n\lambda$$

$$t = \frac{n\lambda}{2\mu} \quad \text{— (1)}$$

For air $\mu = 1$

$$\therefore t = \frac{n\lambda}{2} \quad \text{— (1)'}$$

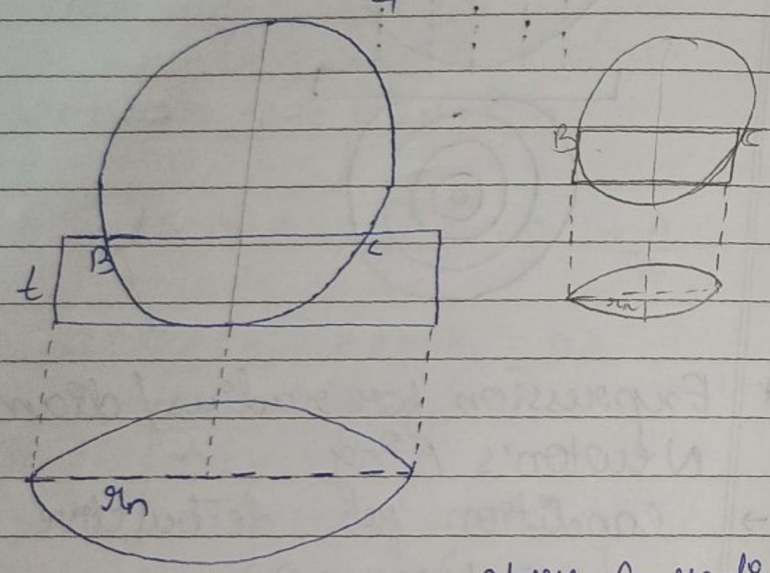
→ Condition for constructive interference

$$2\mu t = (2n+1)\frac{\lambda}{2}$$

$$t = \frac{(2n+1)\lambda}{4\mu} \quad \text{— (II)}$$

For air, $\mu = 1$

$$\therefore t = \frac{(2n+1)\lambda}{4} \quad \text{— (II)'}$$



where R = radius of curvature of plano-convex lens

r_n = radius

In ΔABE

$$AB^2 = BE^2 + AE^2$$

$$R^2 = r_n^2 + (R-t)^2$$

$$R^2 = r_n^2 + R^2 - 2Rt + t^2$$

$$r_n^2 = 2Rt - t^2$$

$$t^2 <<< 2Rt$$

$$\therefore r_n^2 = 2Rt$$

$$r_n^2 = 2 \cdot R \cdot t \quad \text{— (I)}$$

Radius for dark ring

$$r_n^2 = 2R \times \frac{n\lambda}{2\mu}$$

$$= \frac{n\lambda R}{\mu}$$

For air ~~a~~ $\mu = 1$

$$r_n^2 = n\lambda R$$

$$\boxed{r_n = \sqrt{n\lambda R}} \quad n = 0, 1, 2, 3 \ldots$$

$n = 0$

$r_n = 0$

$r_1 = \sqrt{1 \cdot \lambda R}$

$r_2 = \sqrt{2 \cdot \lambda R}$

$r_3 = \sqrt{3\lambda R}$

$r_n \propto \sqrt{n}$

Diameter $D_n = 2r_n$

$$= 2\sqrt{n\lambda R}$$

$$\boxed{D_n = \sqrt{4n\lambda R}}$$

Radius for bright ring

$$r_n^2 = 2 \cdot R \times \frac{(2n+1)\lambda}{4\mu}$$

$$= \frac{\cancel{2}R \times (2n+1)\lambda}{2\mu}$$

Scanned with OKEN Scanner

for air $\mu = 1$

$$r_n^2 = \frac{R \times (2n+1)\lambda}{2 \times 1}$$

$$\boxed{r_n = \sqrt{\frac{R \times (2n+1)\lambda}{2}}} \qquad n = 0, 1, 2, 3, \ldots$$

$$D_n = 2\sqrt{\frac{R \times (2n+1)\lambda}{2}}$$

$$\boxed{D_n = \sqrt{2R(2n+1)\lambda}}$$

$\cancel{r_0 = 0}$ $\quad n = 0 \quad r_0 = \sqrt{\lambda R/2}$

$n = 1 \quad r_1 = \sqrt{\frac{R \cdot 3\lambda}{2}}$

$n = 2 \quad r_2 = \sqrt{\frac{R \cdot 5\lambda}{2}}$

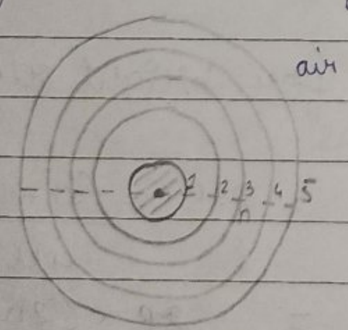
$n = 3 \quad r_3 = \sqrt{\frac{R \cdot 7\lambda}{2}}$

$$r_n \propto \sqrt{\text{odd no.}} = \sqrt{(2n+1)}$$

* Determination of wavelength of monochromatic Light by Newton's Ring Experiment.

p = no. b/w 2 rings

R = radius of curvature of lens (Plano lens) convex



When air film is available

$$r_n^2 = n\lambda R \quad \text{— (I)}$$

$$D_n = 2r_n$$

$$r_n = \frac{D_n}{2}$$

$$r_n^2 = \frac{D_n^2}{4}$$

$$\frac{D_n^2}{4} = n\lambda R$$

$$D_n^2 = 4n\lambda R \quad \text{— (II)}$$

$$r_{n+p}^2 = (n+p)\lambda R \quad \text{— (III)}$$

$$D_{n+p}^2 = 4(n+p)\lambda R \quad \text{— (IV)}$$

eq (IV) − eq (II)

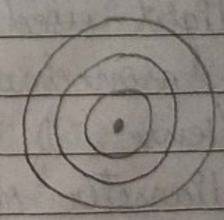
$$D_{n+p}^2 - D_n^2 = 4(n+p)\lambda R - 4n\lambda R$$

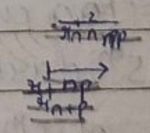
$$= 4n\lambda R + 4p\lambda R - 4n\lambda R$$

$$= 4p\lambda R$$

$$\boxed{\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR}}$$

* ~~Expression for Determination of refractive Index of a liquid by using Newton's Ring Experiment.~~





Q Determine the radius of 10th dark ring and 5th bright ring for an experiment of Newton's ring where monochromatic light of wavelength 5893 Å and plano convex lens of radius of curvature 100cm is used.

→ $\lambda = 5893\ A^\circ = 5893 \times 10^{-10}\ m$

$R = 100\ cm = 10^{-2}\ m$

$r_{10} = ?$ (dark) $n_d = 10$

$r_5 = ?$ (bright) $n_b = 5$

$$r_n = \sqrt{n\lambda R}$$

$$r_{10} = \sqrt{10 \times 5893 \times 10^{-10} \times 10^{-2}}$$

$$= \cancel{1.8635 \times 10^{-3}}\ 2.427 \times 10^{-4}\ m$$

$$r_5 = \sqrt{\frac{R(2n+1)\lambda}{2}}$$

$$= \sqrt{\frac{10^{-2}(11) \times 5893 \times 10^{-10} \times 100}{2}}$$

$$= 1.8003 \times 10^{-3}\ m$$

Q Determine the wavelength of unknown light used in Newton's ring Experiment of ~~monochromatic lens~~ of plano convex lense of $R = 100$ cm in which the diameter of 15th dark ring is 5.36×10^{-3} m and diameter of 5th dark ring is 1.8×10^{-3} m

→ $R = 100\ cm = 10^{-2}\ m$

$D_{15} = 5.36 \times 10^{-3}$ (dark)

$D_5 = 1.8 \times 10^{-3}\ m$ (dark)

$$D_n = 2\sqrt{n\lambda R}$$

$$D_{15} = 2\sqrt{15 \times \lambda \times 10^{-2}}$$

$$5.36 \times 10^{-3} = 2\sqrt{15 \times \lambda \times 10^{-2}}$$

$$\lambda = \frac{D_{n+10}^2 - D_n^2}{4PR}$$

$$= \frac{D_{5+10}^2 - D_5^2}{4(10)(10^{-2})}$$

$$= \frac{D_{15}^2 - D_5^2}{4 \times 10 \times 10^{-2}}$$

$$= \frac{(5.36 \times 10^{-3})^2 - (1.8 \times 10^{-3})^2}{100 \times 40 \times 10^{-2}}$$

$$= \frac{2.872 \times 10^{-5} - 3.24 \times 10^{-6}}{40 \times 10^{-2}}$$

$$= \frac{2.548 \times 10^{-5}}{40 \times 10^{-2}}$$

$$= 6.37 \times 10^{-7}\ m$$

Q In an Newton's ring experiment if the diameter of 20th dark ring is observed to be 6.29×10^{-3} m and the monochromatic light of 5893 Å is used calculate the radius of curvature of plano convex lens

$D_{20} = 6.29 \times 10^{-3} m$

$\lambda = 5893 \text{ A}^\circ = 5893 \times 10^{-10} m$

$R = ?$

$$D_{20} = 2\sqrt{n\lambda R}$$

$$6.29 \times 10^{-3} = 2\sqrt{20 \times 5893 \times 10^{-10} \times R}$$

$$(6.29 \times 10^{-3})^2 = 4(20 \times 5893 \times 10^{-10} \times R)$$

$$3.956 \times 10^{-5} = (4.714 \times 10^{-5})R$$

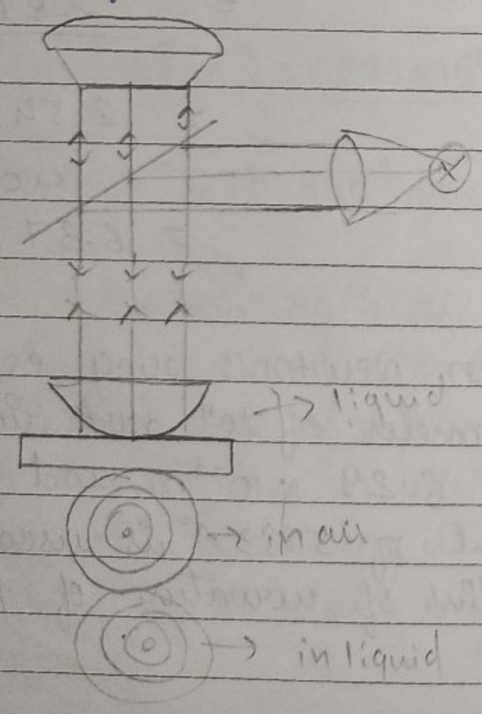
$$R = \frac{3.956 \times 10^{-5}}{4.714 \times 10^{-5}}$$

$$= 0.8392 \, m$$

$$= 83.9 \times 10^{-2} m$$

$$= 83.9 \, cm$$

* Determination of Refractive index of liquid by using Newton's Ring experiment.



Case I : Air film used

$$D_n = \sqrt{4n\lambda R}$$

$$D_n^2 = 4n\lambda R \quad \text{— (I)}$$

$$D_{n+p}^2 = \sqrt{4(n+p)\lambda R}$$

$$= 4(n+p)\lambda R \quad \text{— (II)}$$

$$D_{n+p}^2 - D_n^2 = 4p\lambda R \quad \text{— (III)}$$

Case II : when liquid is used

$$D_n'^2 = \sqrt{\frac{4n\lambda R}{\mu}}$$

$$D_n'^2 = \frac{4n\lambda R}{\mu} \quad \text{— (IV)}$$

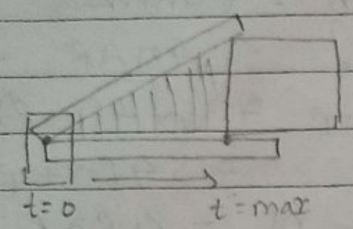
$$(D'_{n+p})^2 = \frac{4(n+p)\lambda R}{\mu} \quad \text{— (V)}$$

$$(D'_{n+p})^2 - D_n'^2 = \frac{4p\lambda R}{\mu} \quad \text{— (VI)}$$

~~from~~ $\frac{\text{eqn (III)}}{\text{eqn VI}}$

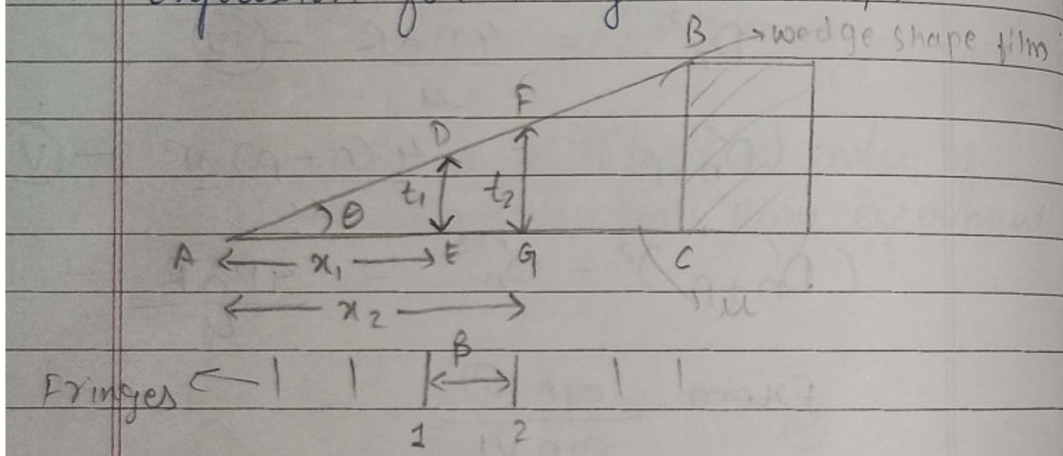
$$\boxed{\frac{D_{n+p}^2 - D_n^2}{D'^2_{n+p} - D'^2_n} = \mu}$$

* Wedge ~~San~~ Shaped Thin film
(non-uniform thickness)



When ends of two glass plates are fixed and another object is kept at another end then their thickness increases.

* Expression for Fringe width (β)



$$x_2 - x_1 = \beta$$

$DE = t_1$ } condition of
$FG = t_2$ } destructive interference

$$t_1 = \frac{n\lambda}{2\mu} \quad -(I)$$

$$t_2 = \frac{(n+1)\lambda}{2\mu} \quad -(II)$$

$$t_2 - t_1 = \frac{(n+1)\lambda}{2\mu} - \frac{n\lambda}{2\mu}$$

$$= \frac{\lambda}{2\mu} \quad -(III)$$

In ΔADE

$$\tan\theta = \frac{DE}{AE} = \frac{t_1}{x_1}$$

$$t_1 = x_1 \tan\theta \quad -(IV)$$

ΔAFG

$$\tan\theta = \frac{FG}{AG} = \frac{t_2}{x_2}$$

$$t_2 = x_2 \tan\theta$$

$$\therefore t_2 - t_1 = (x_2 - x_1)\tan\theta$$

$$= \beta \tan\theta \quad -(V)$$

from (III) & (V)

$$\beta \tan\theta = \frac{\lambda}{2\mu}$$

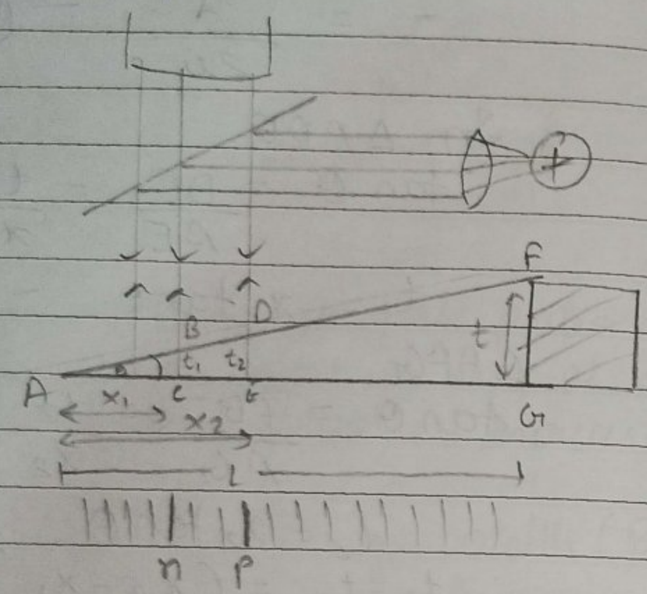
$$\boxed{\beta = \frac{\lambda}{2\mu\tan\theta}}$$

$\theta \rightarrow$ small $\rightarrow$ radian

$\tan\theta = \theta$

$$\therefore \boxed{\beta = \frac{\lambda}{2\mu\theta}}$$

* Expression for determination of thickness of thin paper (t):



$$t_1 = \frac{n\lambda}{2\mu}$$

$$t_2 = \frac{(n+p)\lambda}{2\mu}$$

$$t_2 - t_1 = \frac{(n+p)\lambda}{2\mu} - \frac{n\lambda}{2\mu} = \frac{p\lambda}{2\mu} \quad \text{--- (I)}$$

$$\Delta ABC,\ \tan\theta = \frac{BC}{AC} = \frac{t_1}{x_1}$$

$$\Delta ADE,\ \tan\theta = \frac{DE}{AE} = \frac{t_2}{x_2}$$

$$x_2 - x_1 = \frac{(t_2 - t_1)}{\tan\theta}$$

$$t_2 - t_1 = (x_2 - x_1)\tan\theta \quad \text{--- (II)}$$

From (I) & (2)

$$\frac{p\lambda}{2\mu} = (x_2 - x_1)\tan\theta$$

$$\tan\theta = \frac{p\lambda}{2\mu(x_2 - x_1)} \quad \text{--- (III)}$$

$$\Delta AFG,\ \tan\theta = \frac{FG}{AG} = \frac{t}{L} \quad \text{--- (IV)}$$

From (III) & (IV)

$$\frac{t}{L} = \frac{p\lambda}{2\mu(x_2 - x_1)}$$

$$t = \frac{Lp\lambda}{2\mu(x_2 - x_1)}$$

$$t = \frac{L.\lambda}{2\mu\left(\frac{x_2 - x_1}{p}\right)}$$

$$\boxed{t = \frac{L\lambda}{2\mu\beta}} \quad \text{where } \beta = \frac{x_2 - x_1}{p}$$

Q Determine the thickness of thin paper ~~in~~ by using wedge shaped thin film experiment in which glass plate of length 5 cm is used and light of wavelength 5893 A° is used. The straight line fringes are observed of fringe width 0.235×10^{-3} m.

→ Given

$L = 5\text{cm} = 5 \times 10^{-2}$ m

$\lambda = 5893\ A^\circ = 5893 \times 10^{-10}$ m

$\beta = 0.235 \times 10^{-3}$ m

$$t = \frac{\cancel{\mu\beta}\ L\lambda}{2\mu\beta}$$

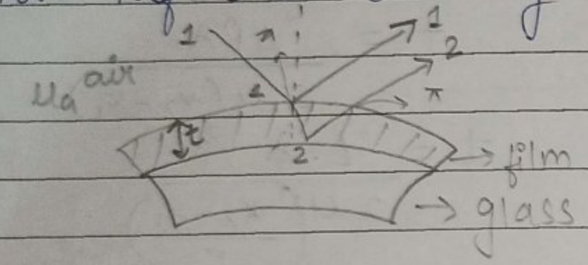
$$= \frac{5\times10^{-2}\times 5893\times10^{-10}}{2\times1\times0.235\times10^{-3}}$$

$$= \cancel{3.462\times10^{-13}}$$

$$= \cancel{6.2691\times10^{-5}\,m}$$

$$= 3.4621\times10^{-12}$$

* Anti-Reflection Coating



$$\mu_f = \sqrt{\mu_g}$$

$$\mu_f < \mu_g$$

$$\mu_a < \mu_f < \mu_g$$

Actual optical path difference

$$= 2\mu_f t\cos r - \frac{\lambda}{2} - \frac{\lambda}{2}$$

$$2\mu_f t\cos r - \frac{\lambda}{2} - \frac{\lambda}{2} = (2n+1)\frac{\lambda}{2}$$

$t = t_{min}$; $n = 0$

at normal incidence $r = 0$; $\cos r = 1$

$\therefore$ Actual optical path diff

$$2\mu_f t - \lambda = [2(0)+1]\frac{\lambda}{2}$$

$$2\mu_f t_{min} - \lambda = \frac{\lambda}{2}$$

$$2\mu_f t_{min} = \frac{\lambda}{2} + \lambda$$

$$\boxed{t_{min} = \frac{\lambda}{4\mu_f}} \approx \frac{\lambda}{2}$$

Q Determine the minimum thickness of anti-reflection coating material which get deposited on the glass surface of refractive index 1.55. If the light of 5500 A° is incident normally

$\Rightarrow$ ~~tm~~ $t_{min} = ?$

$\mu_g = 1.55$

$\lambda = 5500\,A°$

$\mu_f = ?$

$$t_{min} = \frac{\lambda}{4\mu_f}$$

$$\mu_f = \sqrt{\mu_g}$$

$$= \sqrt{1.55}$$

$$= 1.2449$$

$$t_{min} = \frac{5500\ \cancel{\times10^{-10}}}{4\times1.244}$$

1104. $= \cancel{1910.5\times10^{-10}}\ 1104.50$

4.9796